

Skill	Random	Curriculum	Anticurriculum	Skill-stratified	Skill-curriculum	Skill-anticurriculum	SKILL-IT
Answer Verification	2.297 $\pm$ 0.058	2.368 $\pm$ 0.055	2.391 $\pm$ 0.061	2.180 $\pm$ 0.059	2.249 $\pm$ 0.116	2.325 $\pm$ 0.085	2.158 $\pm$ 0.059
Code to Text	0.246 $\pm$ 0.021	0.203 $\pm$ 0.019	1.099 $\pm$ 0.115	0.178 $\pm$ 0.016	0.126 $\pm$ 0.009	1.232 $\pm$ 0.070	0.223 $\pm$ 0.017
Discourse Connective Identification	2.927 $\pm$ 0.069	3.084 $\pm$ 0.067	2.932 $\pm$ 0.058	2.805 $\pm$ 0.071	2.891 $\pm$ 0.001	2.925 $\pm$ 0.011	2.784 $\pm$ 0.068
Entity Generation	2.033 $\pm$ 0.421	2.012 $\pm$ 0.437	2.363 $\pm$ 0.234	1.803 $\pm$ 0.384	1.853 $\pm$ 0.483	2.068 $\pm$ 0.719	1.863 $\pm$ 0.418
Entity Relation Classification	1.020 $\pm$ 0.147	1.014 $\pm$ 0.140	1.533 $\pm$ 0.138	0.859 $\pm$ 0.131	0.825 $\pm$ 0.022	0.959 $\pm$ 0.009	0.908 $\pm$ 0.146
Information Extraction	2.154 $\pm$ 0.040	2.247 $\pm$ 0.037	2.352 $\pm$ 0.042	2.140 $\pm$ 0.037	2.286 $\pm$ 0.022	2.338 $\pm$ 0.025	2.073 $\pm$ 0.042
Irony Detection	3.024 $\pm$ 0.154	3.798 $\pm$ 0.095	2.942 $\pm$ 0.158	2.680 $\pm$ 0.146	3.889 $\pm$ 0.066	2.099 $\pm$ 0.152	2.797 $\pm$ 0.155
Preposition Prediction	0.979 $\pm$ 0.124	0.887 $\pm$ 0.147	1.488 $\pm$ 0.213	0.845 $\pm$ 0.152	0.941 $\pm$ 0.019	1.044 $\pm$ 0.029	0.876 $\pm$ 0.173
Punctuation Error Detection	2.950 $\pm$ 0.065	3.120 $\pm$ 0.052	2.961 $\pm$ 0.064	3.264 $\pm$ 0.061	3.019 $\pm$ 0.010	3.360 $\pm$ 0.013	3.216 $\pm$ 0.055
Question Answering	2.277 $\pm$ 0.005	2.367 $\pm$ 0.006	2.398 $\pm$ 0.006	2.542 $\pm$ 0.004	2.689 $\pm$ 0.001	2.707 $\pm$ 0.016	2.448 $\pm$ 0.008
Question Generation	2.617 $\pm$ 0.005	2.777 $\pm$ 0.015	2.695 $\pm$ 0.008	2.783 $\pm$ 0.021	3.062 $\pm$ 0.006	2.876 $\pm$ 0.032	2.666 $\pm$ 0.012
Question Understanding	1.965 $\pm$ 0.051	2.199 $\pm$ 0.059	2.060 $\pm$ 0.033	1.958 $\pm$ 0.051	2.385 $\pm$ 0.022	2.100 $\pm$ 0.054	1.895 $\pm$ 0.043
Sentence Expansion	2.501 $\pm$ 0.095	2.598 $\pm$ 0.097	2.583 $\pm$ 0.074	2.225 $\pm$ 0.095	2.311 $\pm$ 0.076	2.408 $\pm$ 0.074	2.236 $\pm$ 0.083
Sentiment Analysis	3.203 $\pm$ 0.012	3.415 $\pm$ 0.016	3.209 $\pm$ 0.010	3.278 $\pm$ 0.014	3.607 $\pm$ 0.012	3.308 $\pm$ 0.015	3.213 $\pm$ 0.012
Stance Detection	1.810 $\pm$ 0.100	1.775 $\pm$ 0.120	2.231 $\pm$ 0.128	1.385 $\pm$ 0.070	1.361 $\pm$ 0.114	1.823 $\pm$ 0.189	1.556 $\pm$ 0.125
Summarization	2.961 $\pm$ 0.015	3.149 $\pm$ 0.023	3.041 $\pm$ 0.014	2.960 $\pm$ 0.019	3.323 $\pm$ 0.028	3.021 $\pm$ 0.013	2.907 $\pm$ 0.012
Text Categorization	2.488 $\pm$ 0.023	2.692 $\pm$ 0.029	2.553 $\pm$ 0.006	2.570 $\pm$ 0.015	3.001 $\pm$ 0.007	2.635 $\pm$ 0.014	2.448 $\pm$ 0.017
Text Matching	2.177 $\pm$ 0.059	2.232 $\pm$ 0.055	2.316 $\pm$ 0.048	2.152 $\pm$ 0.061	2.324 $\pm$ 0.004	2.304 $\pm$ 0.035	2.093 $\pm$ 0.054
Text Simplification	2.155 $\pm$ 0.023	2.193 $\pm$ 0.039	2.325 $\pm$ 0.033	1.926 $\pm$ 0.026	2.037 $\pm$ 0.005	2.156 $\pm$ 0.011	1.952 $\pm$ 0.026
Text to Code	0.560 $\pm$ 0.037	0.495 $\pm$ 0.036	1.215 $\pm$ 0.052	0.490 $\pm$ 0.029	0.433 $\pm$ 0.014	1.455 $\pm$ 0.086	0.553 $\pm$ 0.042
Toxic Language Detection	3.106 $\pm$ 0.027	3.496 $\pm$ 0.017	3.058 $\pm$ 0.029	3.199 $\pm$ 0.024	3.758 $\pm$ 0.025	3.155 $\pm$ 0.050	3.129 $\pm$ 0.020
Word Semantics	2.092 $\pm$ 0.027	2.334 $\pm$ 0.034	2.156 $\pm$ 0.064	1.916 $\pm$ 0.043	1.784 $\pm$ 0.048	2.424 $\pm$ 0.038	1.952 $\pm$ 0.019
Wrong Candidate Generation	2.438 $\pm$ 0.021	2.606 $\pm$ 0.039	2.519 $\pm$ 0.027	2.506 $\pm$ 0.026	2.849 $\pm$ 0.029	2.574 $\pm$ 0.018	2.432 $\pm$ 0.025
Average	2.173 $\pm$ 0.028	2.307 $\pm$ 0.025	2.366 $\pm$ 0.026	2.115 $\pm$ 0.027	2.304 $\pm$ 0.031	2.317 $\pm$ 0.052	2.103 $\pm$ 0.032

Table 7: Validation loss per skill for data selection in continual pre-training setting on a subset of the Natural Instructions Dataset.

Out-of-domain In Table 8, we provide a breakdown of validation loss per evaluation skill under random sampling on the training data, skill-stratified sampling over prerequisite skills (e.g., the nonzero rows in Figure 14), and SKILL-IT.

Skill	Random	Skill-stratified	SKILL-IT
Answerability Classification	3.048 $\pm$ 0.003	3.076 $\pm$ 0.002	3.043 $\pm$ 0.003
Cause Effect Classification	2.068 $\pm$ 0.004	2.101 $\pm$ 0.005	2.067 $\pm$ 0.006
Coreference Resolution	3.101 $\pm$ 0.003	3.142 $\pm$ 0.004	3.099 $\pm$ 0.004
Data to Text	2.363 $\pm$ 0.004	2.388 $\pm$ 0.005	2.359 $\pm$ 0.005
Dialogue Act Recognition	2.329 $\pm$ 0.009	2.364 $\pm$ 0.010	2.320 $\pm$ 0.009
Grammar Error Correction	2.399 $\pm$ 0.008	2.418 $\pm$ 0.009	2.389 $\pm$ 0.007
Keyword Tagging	2.744 $\pm$ 0.005	2.760 $\pm$ 0.007	2.733 $\pm$ 0.006
Overlap Extraction	2.749 $\pm$ 0.011	2.763 $\pm$ 0.012	2.733 $\pm$ 0.010
Question Rewriting	2.591 $\pm$ 0.009	2.628 $\pm$ 0.011	2.586 $\pm$ 0.010
Textual Entailment	2.472 $\pm$ 0.002	2.503 $\pm$ 0.003	2.468 $\pm$ 0.002
Title Generation	3.027 $\pm$ 0.002	3.037 $\pm$ 0.002	3.015 $\pm$ 0.002
Word Analogy	1.665 $\pm$ 0.016	1.682 $\pm$ 0.015	1.668 $\pm$ 0.016
Average	2.546 $\pm$ 0.003	2.572 $\pm$ 0.003	2.540 $\pm$ 0.003

Table 8: Validation loss per skill for data selection in out-of-domain setting over Natural Instructions train task split and test task split.

In Table 9 we provide a breakdown of the RedPajama experiment’s accuracy per evaluation skill, corresponding to the results in Figure 7.

	1 Billion Tokens		2 Billion Tokens		3 Billion Tokens	
	Uniform	SKILL-IT	Uniform	SKILL-IT	Uniform	SKILL-IT
ARC Challenge (acc norm)	35.4	34.6	35.3	34.9	34.6	34.8
ARC Easy (acc norm)	62.2	61.2	62.4	61.7	62.5	62.0
BoolQ	68.9	68.2	67.7	68.6	67.2	68.7
COPA	81.0	82.0	80.0	81.0	81.0	81.0
HellaSwag (acc norm)	63.9	63.7	63.8	63.9	64.0	63.9
LAMBADA OpenAI	64.4	67.0	65.9	66.7	66.8	66.0
PIQA (acc norm)	74.8	75.0	75.5	75.2	75.0	75.7
Winogrande	62.8	63.9	63.9	63.2	63.4	63.1
Average accuracy	64.2	64.4	64.3	64.4	64.3	64.4

Table 9: Performance of model trained on RedPajama with uniform sampling and SKILL-IT on LM evaluation harness. Unless otherwise noted, accuracy is reported for each task.

D.3.2 Weight trajectories

We provide SKILL-IT’s weight trajectories for each result. The weight per skill across training steps for the LEGO pre-training experiment corresponding to Figure 4 (left) is shown in Figure 20. We see that SKILL-IT initially allocates more weight to skill 2 and less to 1, 3, 4, 5. Since skill 1 is learned quickly, the weight on skill 1 immediately drops to below 0.1 at 1000 steps. The weight on skills 3, 4, and 5 increase from around 0 to 3000 steps, during which their respective validation losses are higher than those of skills 1 and 2. Near the end of training, all losses are converging to 0, and so the weight per skill is roughly uniform.

The weight per skill across training steps for the addition pre-training experiment corresponding to Figure 4 (right) is shown in Figure 21. SKILL-IT allocates more weight to skill 2, which has an edge to skill 1 as shown in Figure 11. It also allocates very little weight to skill 3, which is learned faster than the other two skills. Eventually, it puts more weight on skill 1, the hardest skill, and then converges to uniform sampling as all validation losses approach 0.

The weight per skill across training steps for the LEGO fine-tuning experiment and the Spanish question generation and stance detection experiments corresponding to Figure 5 is shown in Figure 22. Since there is only one target skill in these experiments, the mixture of weights approaches uniform as the loss on the target skill approaches 0. It is interesting

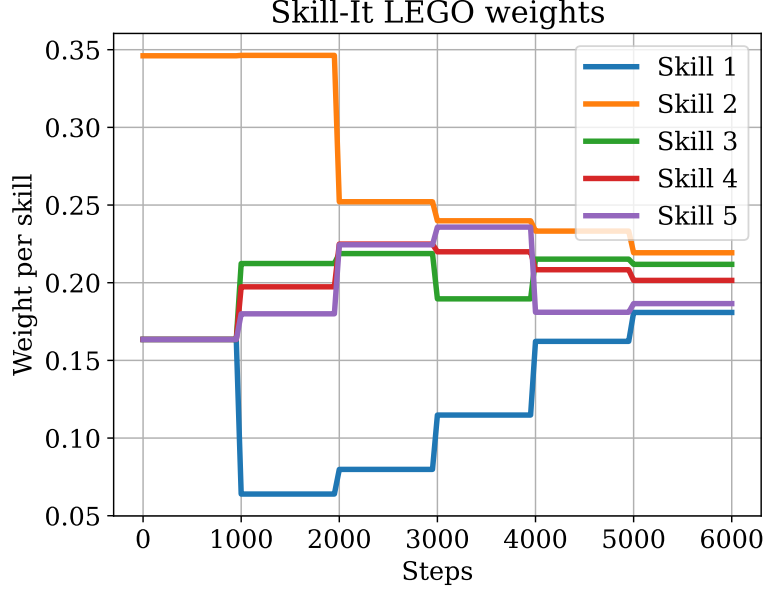


Figure 20: Weight per skill for LEGO pre-training experiment. SKILL-IT initially allocates more weight to skill 2, but eventually puts more weight on harder skills (3, 4, 5) before converging to uniform sampling when all losses converge roughly to 0.

to explore how to reduce edge weights and regularization so that the mixture approaches the target skill instead, although preliminary experiments where we decayed the edge weight and the strength of the Bregman divergence term did not appear better. We hypothesize that since training on a uniform mixture (as in Figure 3) did strictly better than training on the target skill and their loss curves did not intersect during the training run, it is better to allocate non-negligible weight on all skills throughout the training run.

The weight per skill across training steps for the Natural Instructions out-of-domain experiment corresponding to Figure 6 is shown in Figure 23, where the legend is provided for the top 10 task categories with the largest weights. While the initial weights based on the skills graph roughly establishes the order of weight magnitude, the differences among the losses on the evaluation skills increases the range of weights as training continues. As validation losses saturate, the weights also converge to fixed values.

D.4 Experiments on 1.3B parameter model

We demonstrate that the skills graph learned on the 125M parameter model can be used for data selection with the GPT-Neo-1.3B model. We present results in the continual pre-training setting on the LEGO synthetic and Natural Instructions.

All results are reported over 3 random seeds. For the LEGO experiment, we train for 1500 steps with $\eta = 0.5$, $T = 30$, $w = 3$. For the NI experiment, we train for 5000 steps with $\eta = 0.2$, and $T = 1$. The skill graphs were learned using the 125M parameter model as described in section C.2.

In Figure 24, we train the 1.3B model using SKILL-IT for the LEGO synthetic and find that it still outperforms random and skill-stratified sampling on average. In particular, while performance across sampling methods is similar for early skills, the discrepancy is larger for skill 5, for which SKILL-IT allocates more weight to dynamically. In Figure 25, we provide the weight trajectories of SKILL-IT. We observe that the weight trajectories are similar to that on the 125M parameter model, where initial weight is allocated towards skill 2. Later on, more weight is allocated towards skills 4 and 5, whose losses are higher, and eventually the weight mixture converges to uniform as all losses converge to near 0.

In Table 10, we report performance of SKILL-IT with the 1.3B model on the Natural Instructions pre-training experiment and find that the trends from the smaller model hold—SKILL-IT outperforms random and skill-stratified sampling on average.

D.5 Ablations

We report ablations on the skills graph and the online component of SKILL-IT. Instead of using A in Algorithm 1, we study the performance when the identity matrix is used instead; intuitively, this corresponds to a misspecified skills graph where